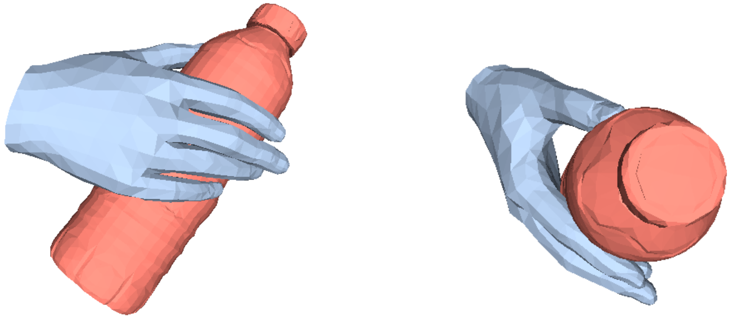
Lecture 8B: Visual Imitation

Jitendra Malik

Imitation Learning for Robot Manipulation

image



3D reconstruction

Reconstructing Hand-Object Interactions in the Wild
Zhe Cao, Ilija Radosavovic, Angjoo Kanazawa, Jitendra Malik (ICCV 2021)

Reconstructing Hands in 3D with Transformers

Georgios Pavlakos, Dandan Shan, Ilija Radosavovic, Angjoo Kanazawa, David Fouhey, Jitendra Malik

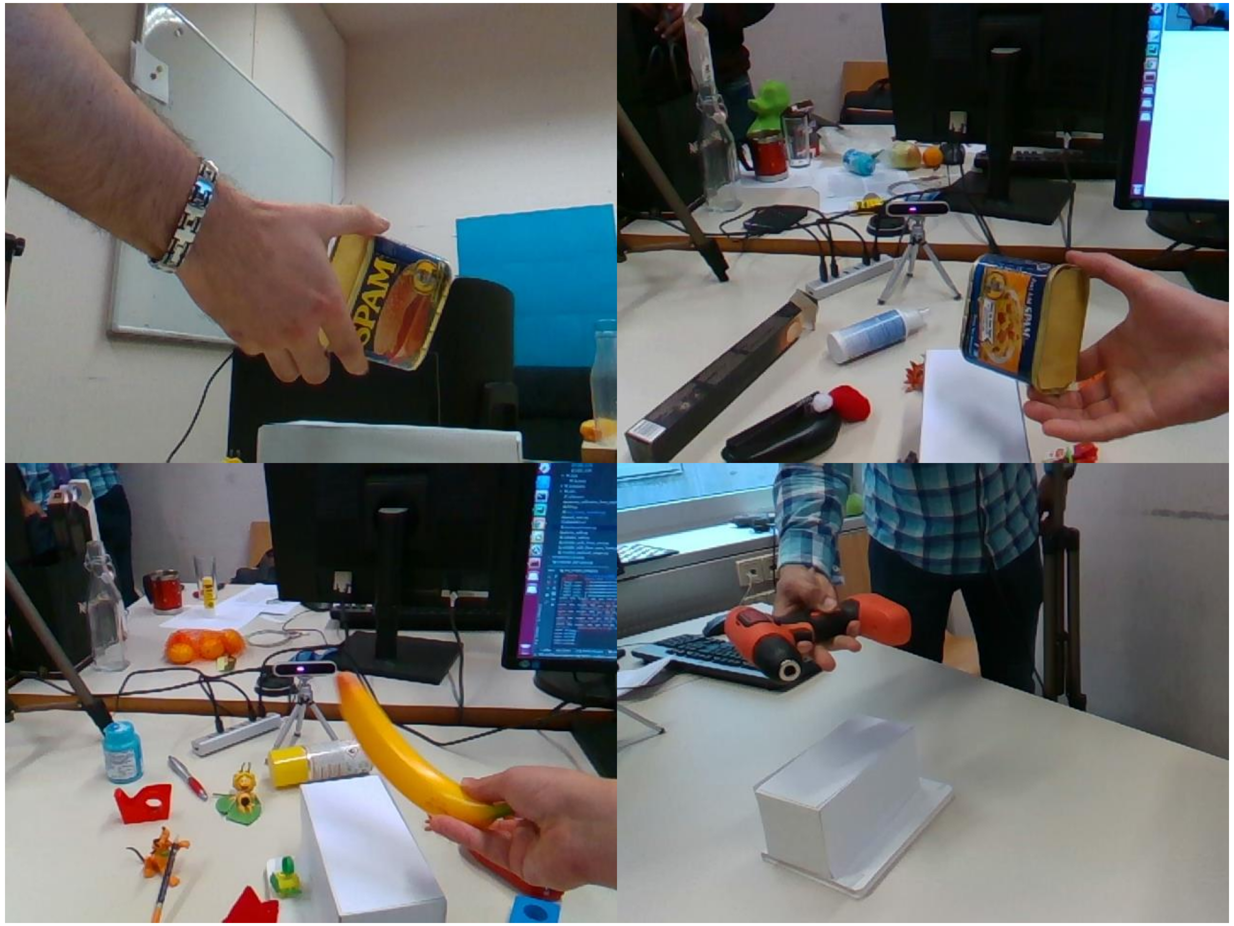

HaMeR Results
Contact Juggling

Evaluation - FreiHAND

3D Pose / 3D Mesh Accuracy

Method	PA-MPJPE ↓	PA-MPVPE ↓	F@5 ↑	F@15 ↑
I2L-MeshNet [39]	7.4	7.6	0.681	0.973
Pose2Mesh [10]	7.7	7.8	0.674	0.969
I2UV-HandNet [7]	6.7	6.9	0.707	0.977
METRO [33]	6.5	6.3	0.731	0.984
Tang *et al.* [53]	6.7	6.7	0.724	0.981
Mesh Graphormer [34]	5.9	6.0	0.764	0.986
MobRecon [8]	**5.7**	5.8	0.784	0.986
AMVUR [24]	6.2	6.1	0.767	0.987
Ours	6.0	**5.7**	**0.785**	**0.990**

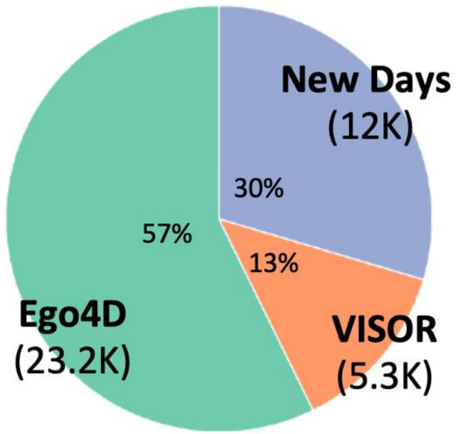
Evaluation - HO3D



3D Pose / 3D Mesh Accuracy

Method	AUC_J ↑	PA-MPJPE ↓	AUC_V ↑	PA-MPVPE ↓	F@5 ↑	F@15 ↑
Liu *et al.* [36]	0.803	9.9	0.810	9.5	0.528	0.956
HandOccNet [44]	0.819	9.1	0.819	8.8	0.564	0.963
I2UV-HandNet [7]	0.804	9.9	0.799	10.1	0.500	0.943
Hampali *et al.* [19]	0.788	10.7	0.790	10.6	0.506	0.942
Hasson *et al.* [21]	0.780	11.0	0.777	11.2	0.464	0.939
ArtiBoost [58]	0.773	11.4	0.782	10.9	0.488	0.944
Pose2Mesh [10]	0.754	12.5	0.749	12.7	0.441	0.909
I2L-MeshNet [39]	0.775	11.2	0.722	13.9	0.409	0.932
METRO [33]	0.792	10.4	0.779	11.1	0.484	0.946
MobRecon[8]	-	9.2	-	9.4	0.538	0.957
Keypoint Trans [20]	0.786	10.8	-	-	-	-
AMVUR [24]	0.835	8.3	0.836	8.2	0.608	0.965
Ours	**0.846**	**7.7**	**0.841**	**7.9**	**0.635**	**0.980**

Hint - **H**and **Int**eractions Dataset



New Days of Hands

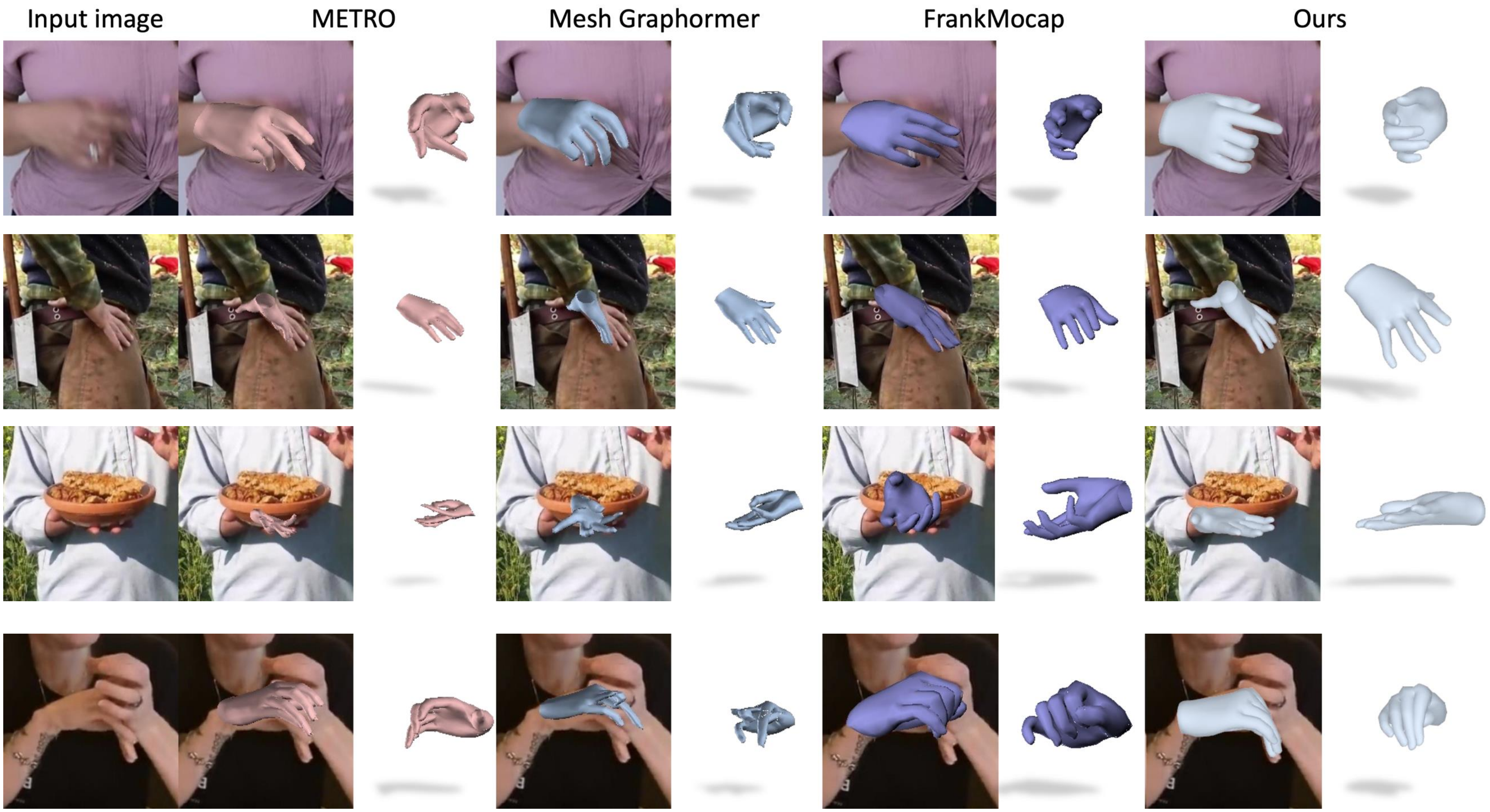
Epic Kitchens

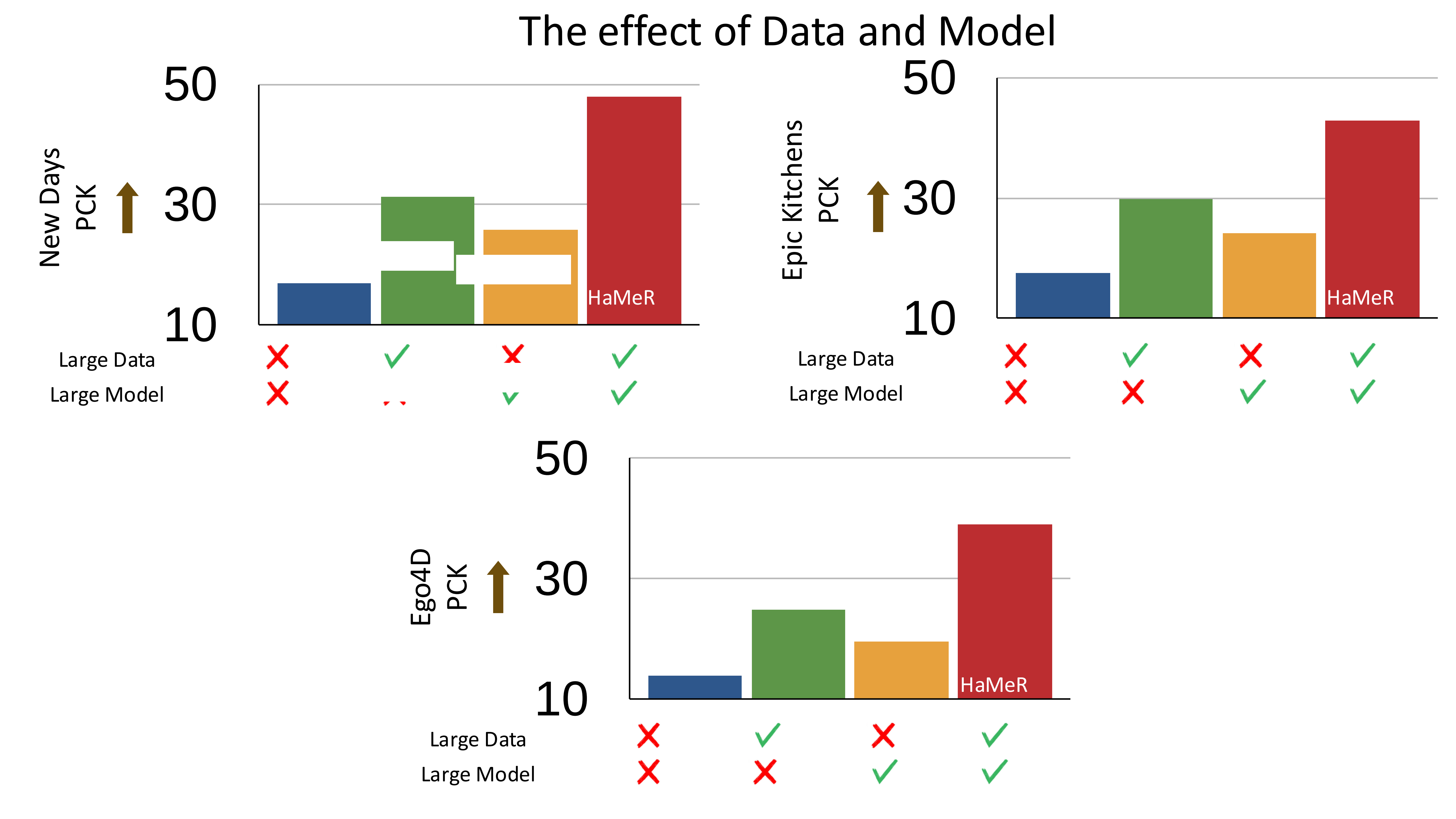
Ego4D

Evaluation - HInt

	Method	New Days @0.05	@0.1	@0.15
All Joints	FrankMocap [49]	16.1	41.4	60.2
	METRO [33]	14.7	38.8	57.3
	MeshGraphormer [34]	16.8	42.0	59.7
	HandOccNet (param) [44]	9.1	28.4	47.8
	HandOccNet (no param) [44]	13.7	39.1	59.3
	Ours	**48.0**	**78.0**	**88.8**

Comparisons with the State of the Art





The effect of Data and Model
New Days
PCK
50
30
10
HaMeR
Large Data
Large Model
Epic Kitchens
PCK
50
30
10
HaMeR
Large Data
Large Model
Ego4D
PCK
50
30
10
HaMeR
Large Data
Large Model

Reconstructing Hand-Held Objects in 3D

Jane Wu, Georgios Pavlakos, Georgia Gkioxari, Jitendra Malik

https://arxiv.org/abs/2404.06507

Builds on two previous foundation models : HaMeR (Berkeley) and MCC (Meta)

Multiview Compressive Coding for 3D Reconstruction

CY Wu, J Johnson, J Malik, C Feichtenhofer, G. Gkioxari

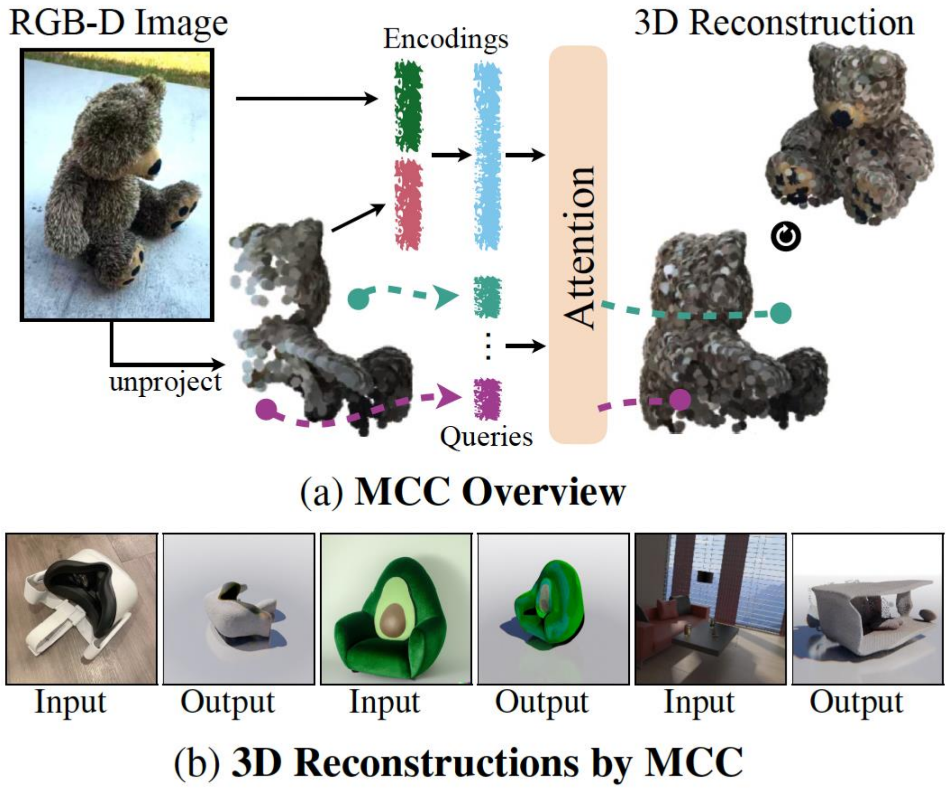
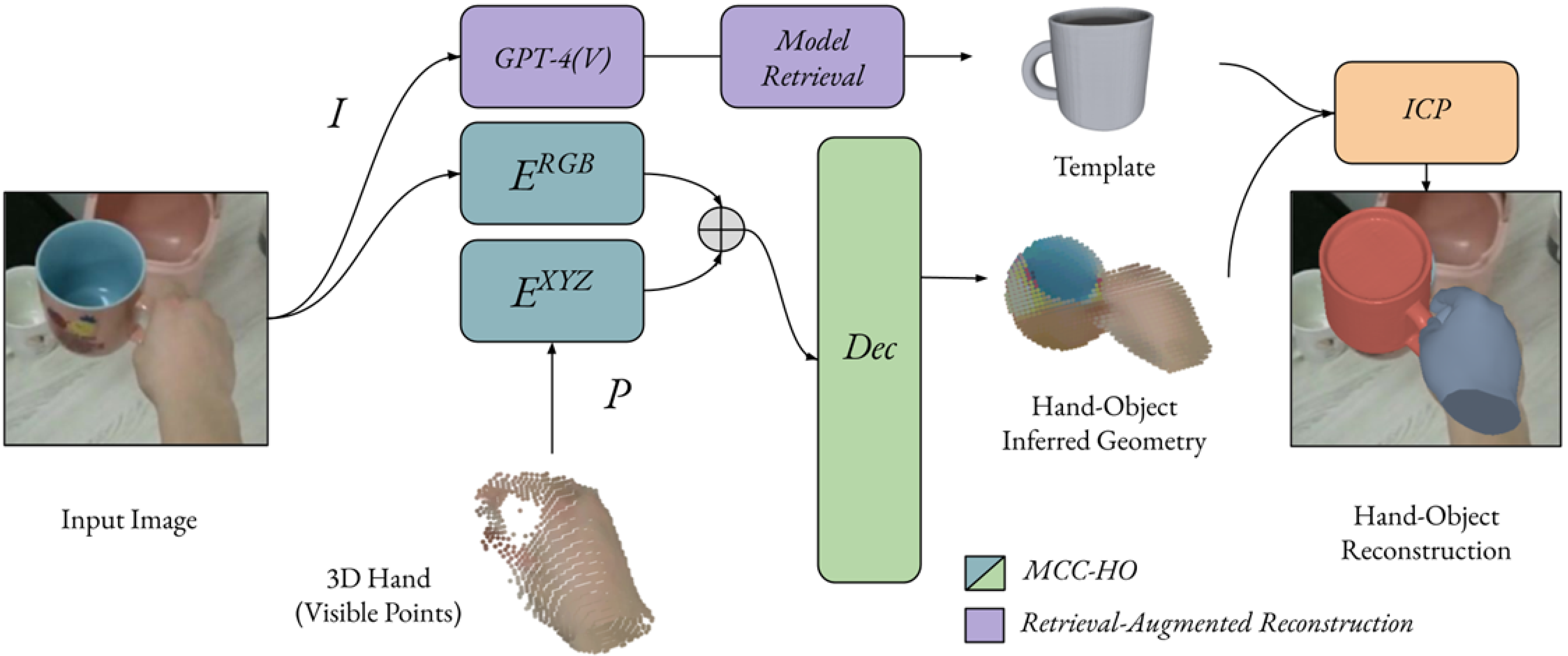


Figure 1. **Multiview Compressive Coding (MCC).** (a): MCC encodes an input RGB-D image and uses an attention-based model to predict the occupancy and color of query points to form the final 3D reconstruction. (b): MCC generalizes to novel objects captured with iPhones (left) or imagined by DALL·E 2 [47] (middle). It is also general – it works not only on objects but also scenes (right).



GPT-4(V)
Model Retrieval
Template
ICP
I
E^{RGB}
E^{XYZ}
Dec
P
Input Image
3D Hand (Visible Points)
Hand-Object Inferred Geometry
Hand-Object Reconstruction
MCC-HO
Retrieval-Augmented Reconstruction

Reconstruction

MCC-HO Results

	DexYCB			MOW			HOI4D		
	F-5 (↑)	F-10 (↑)	CD (↓)	F-5 (↑)	F-10 (↑)	CD (↓)	F-5 (↑)	F-10 (↑)	CD (↓)
HO [24]	0.24	0.48	4.76	0.03	0.06	49.8	0.28	0.51	3.86
IHOI [90]	-	-	-	0.13	0.24	23.1	0.42	0.70	2.7
MCC-HO	**0.36**	**0.60**	**3.74**	**0.15**	**0.31**	**15.2**	**0.52**	**0.78**	**1.36**

Comparison to prior work. Chamfer Distance (cm^2) and F-score (5mm, 10mm).




ACTIVATOR SMOOTHIES
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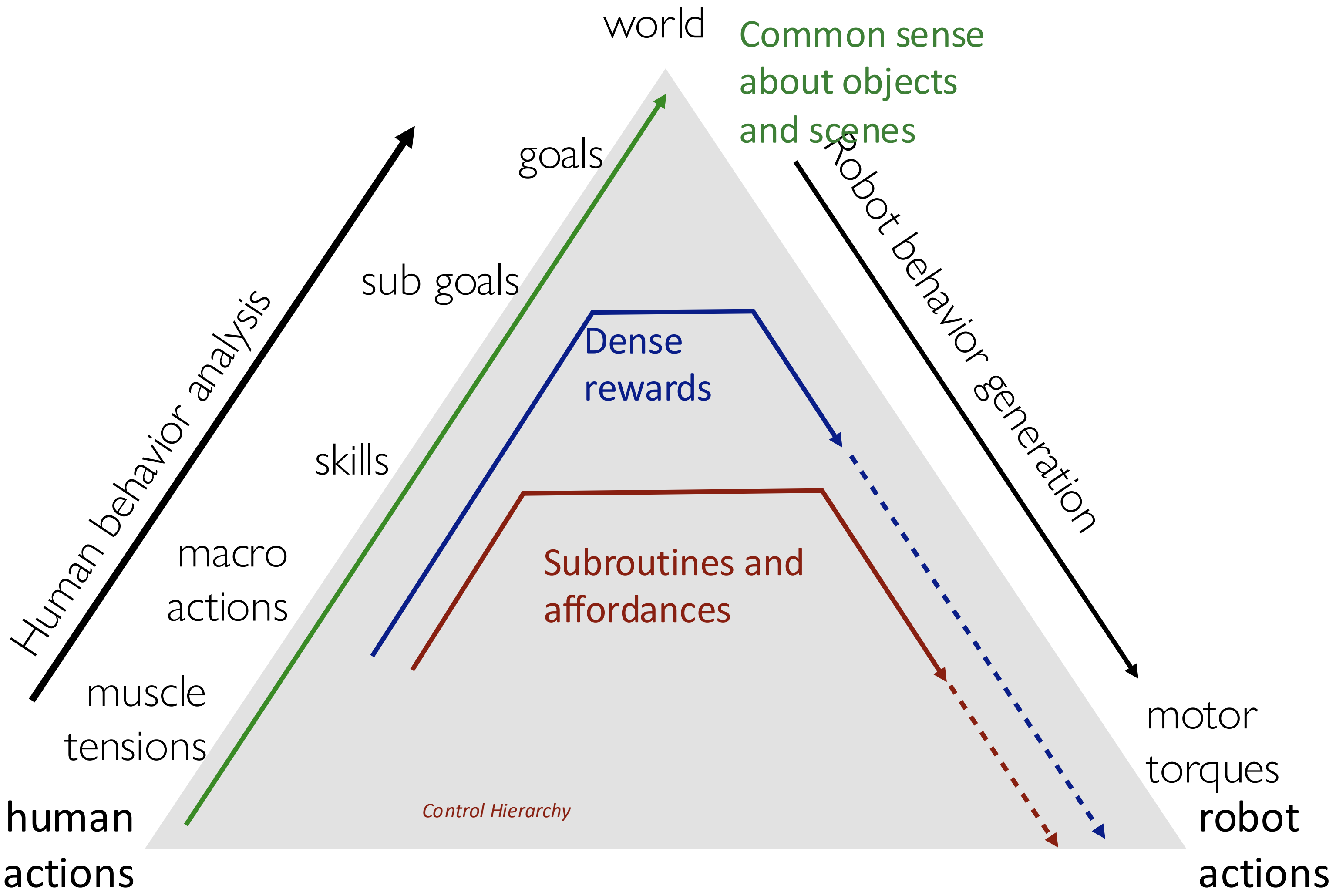
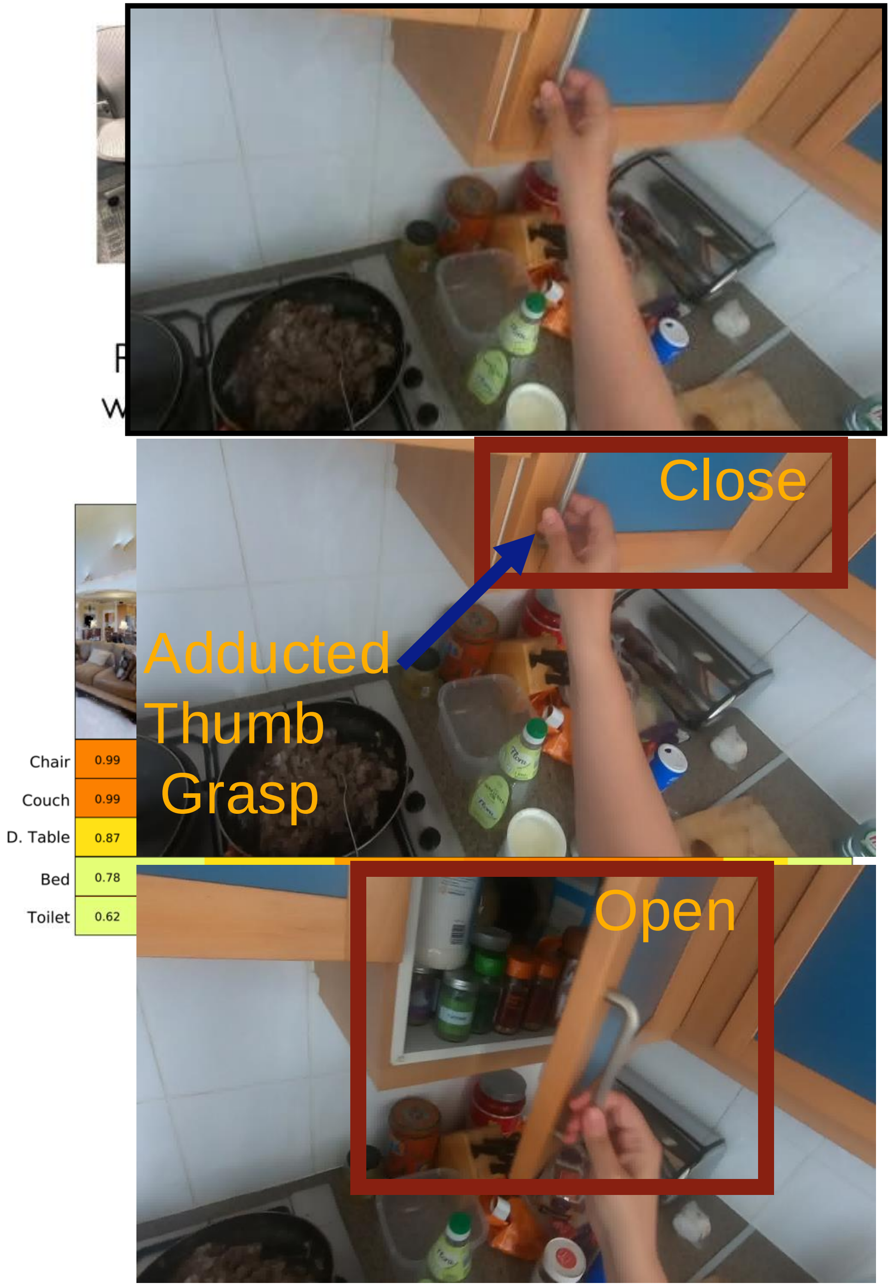
Campus


Transferring at different abstraction levels

Credit: Saurabh Gupta, UIUC





Interactive Object Understanding

A) Which sites can we interact at?
(cupboard handles)

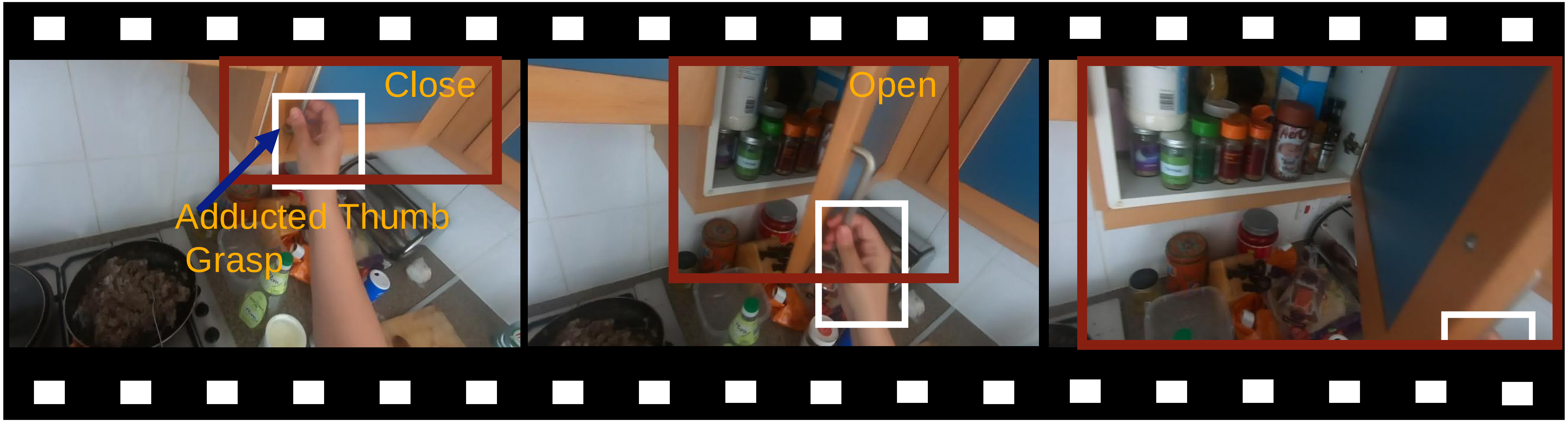
B) How to interact with those sites?
(using adducted thumb grasp)

C) What happens when we do?
(the cupboard opens to reveal condiments)

Learn through observation of ***human hands*** *interacting with the world.*

Goyal et. al. Human Hands as Probes for Interactive Object Understanding. CVPR 2022

Human Hands in Egocentric Videos are Informative



1. Attending to hands localizes and stabilizes active objects.
2. Hands show where all we can interact in the scene.
3. Analyzing hands reveals information about objects: their state and how to interact with them.

Slide credit: Saurabh Gupta

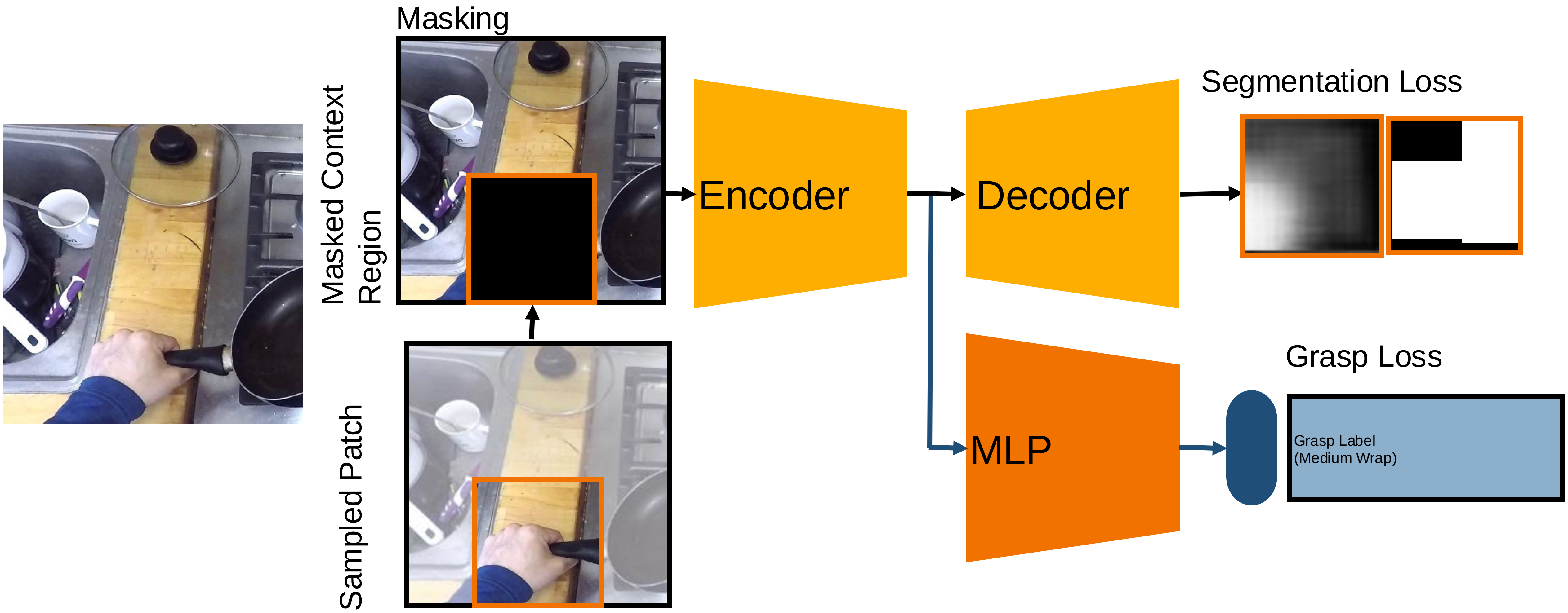
A,B) Learning Object Affordances

Data creation using off-the-shelf models

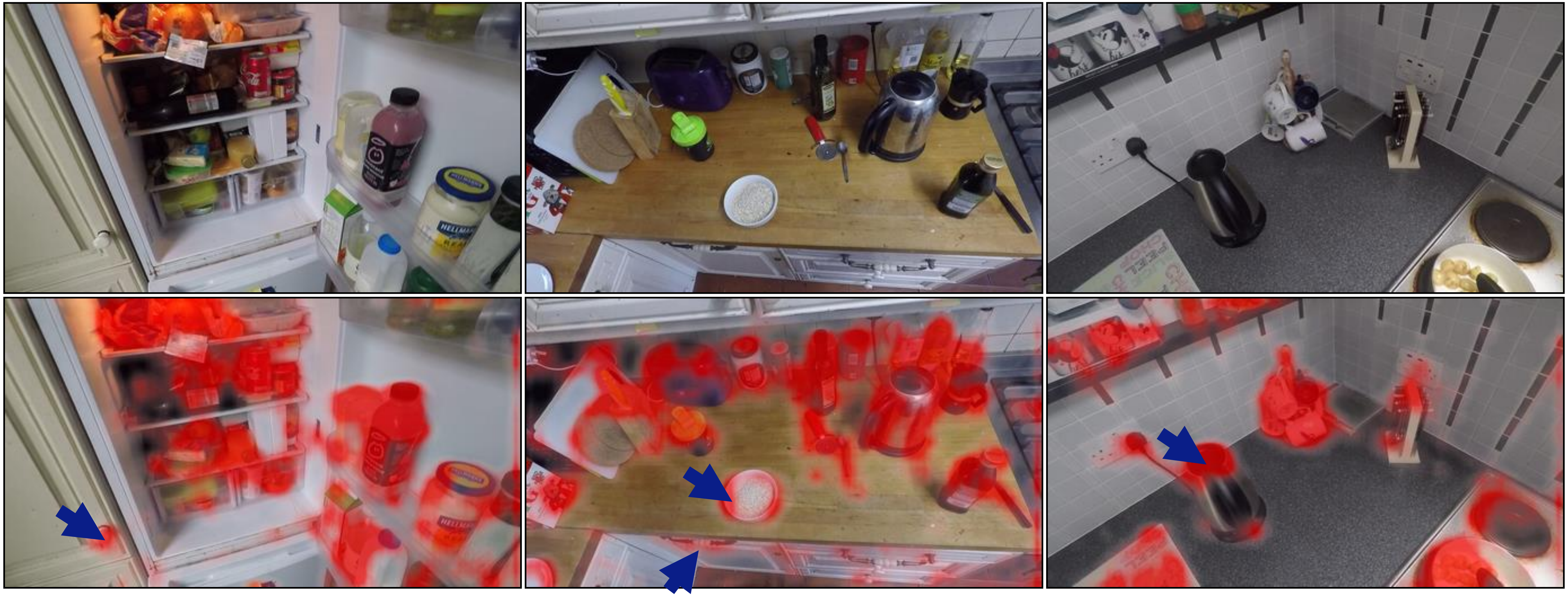
Shan et al. CVPR 2020. Understanding Human Hands in Contact at Internet

A,B) Learning Object Affordances

Affordance from Context Prediction (ACP)



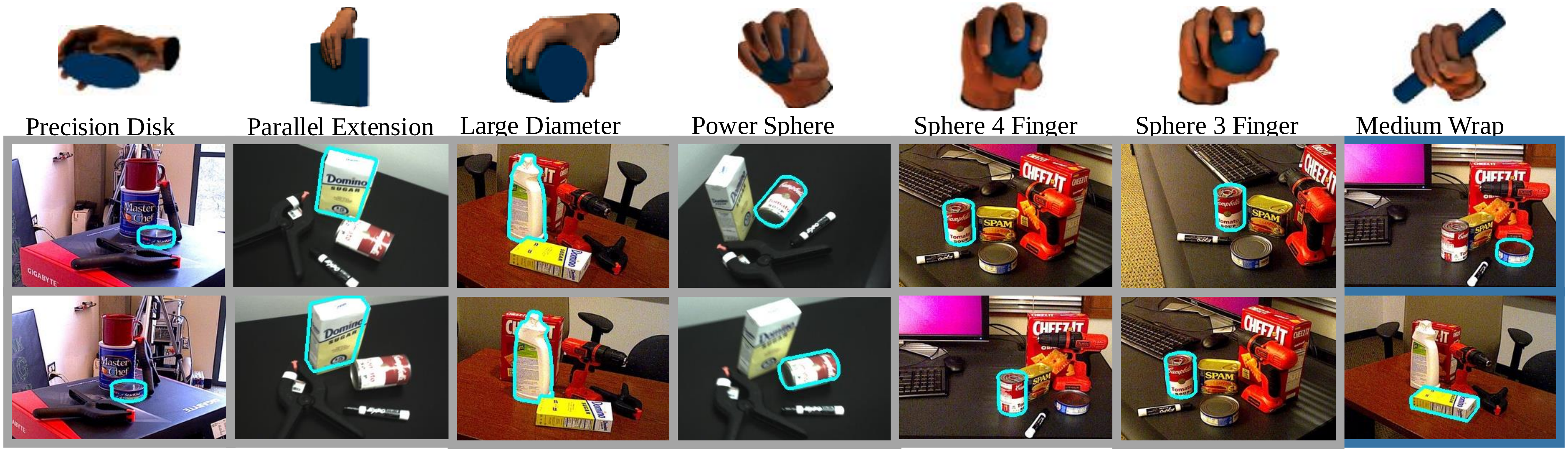
Learning Object Affordances: Results



Learning Object Affordances: Results

Grasps Afforded by Objects (GAO) Task

Top Ranking Objects predicted by ACP



Interactive Object Understanding

A) Which sites can we interact at?
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C) What happens when we do?
(the cupboard opens to reveal condiments)

*Learn through observation of **human hands** interacting with the world.*

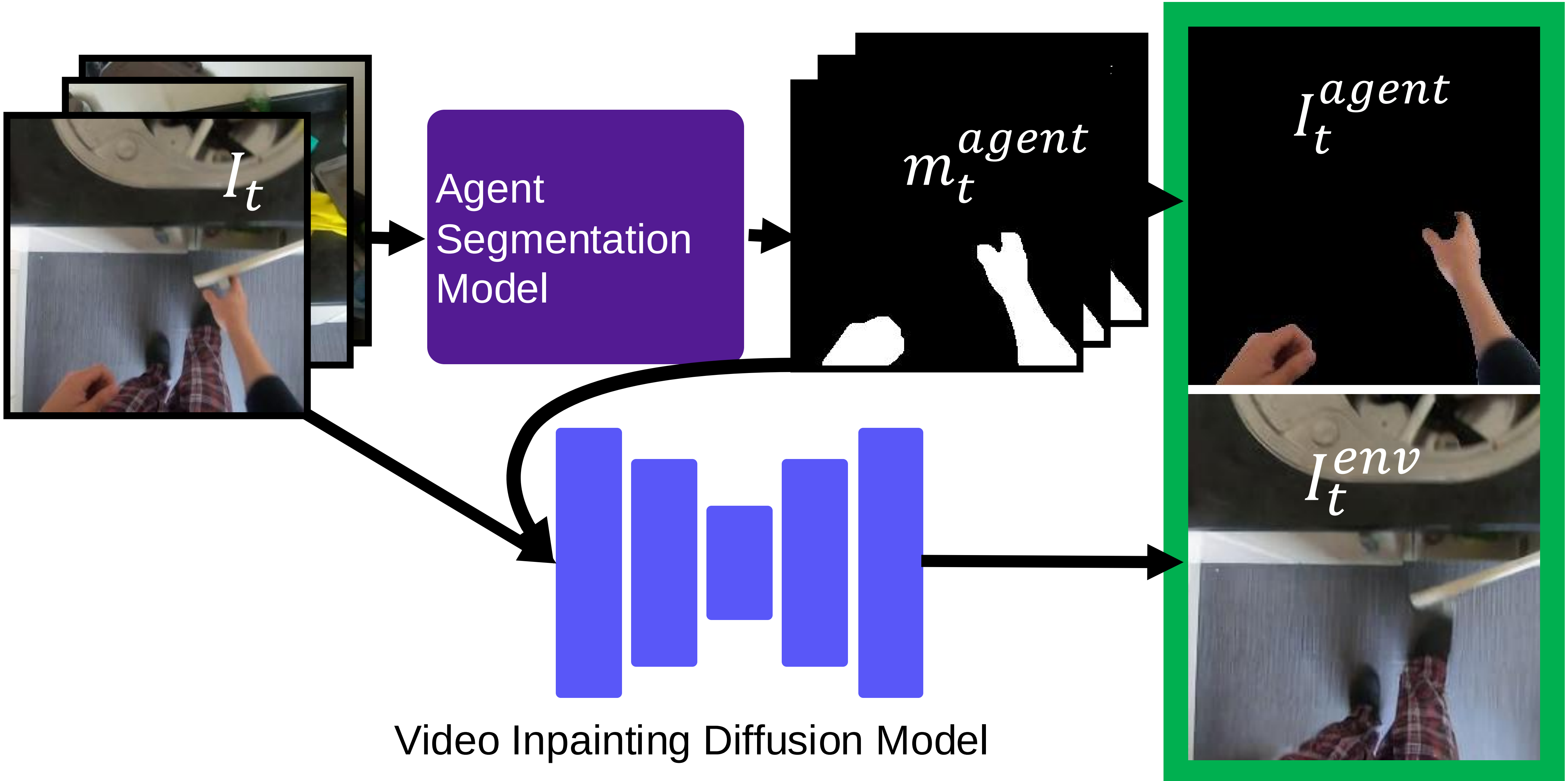
Slide credit: Saurabh Gupta

Look Ma, No Hands! Agent-Environment Factorization of Egocentric Videos

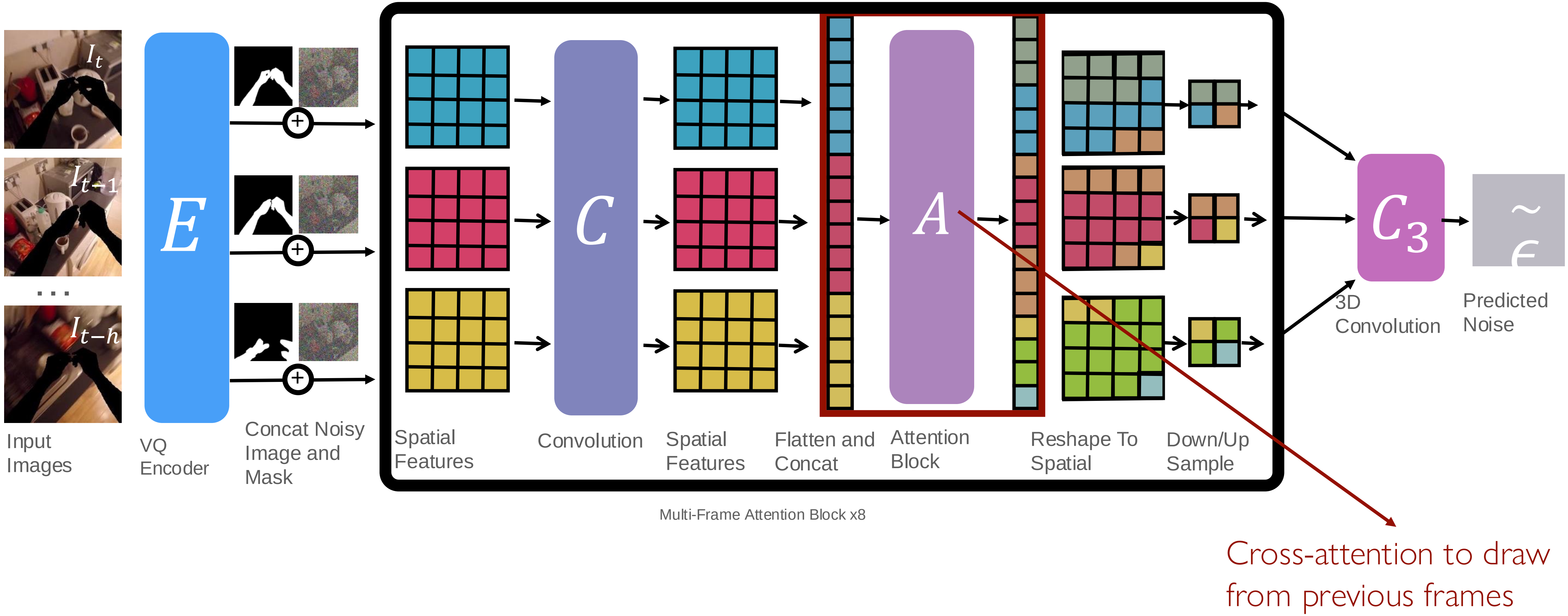
- Matthew Chang, Aditya Prakash, Saurabh Gupta

NeurIPS 2023

Agent-Environment *Factored* Representations



Video Inpainting Diffusion Model (VIDM)

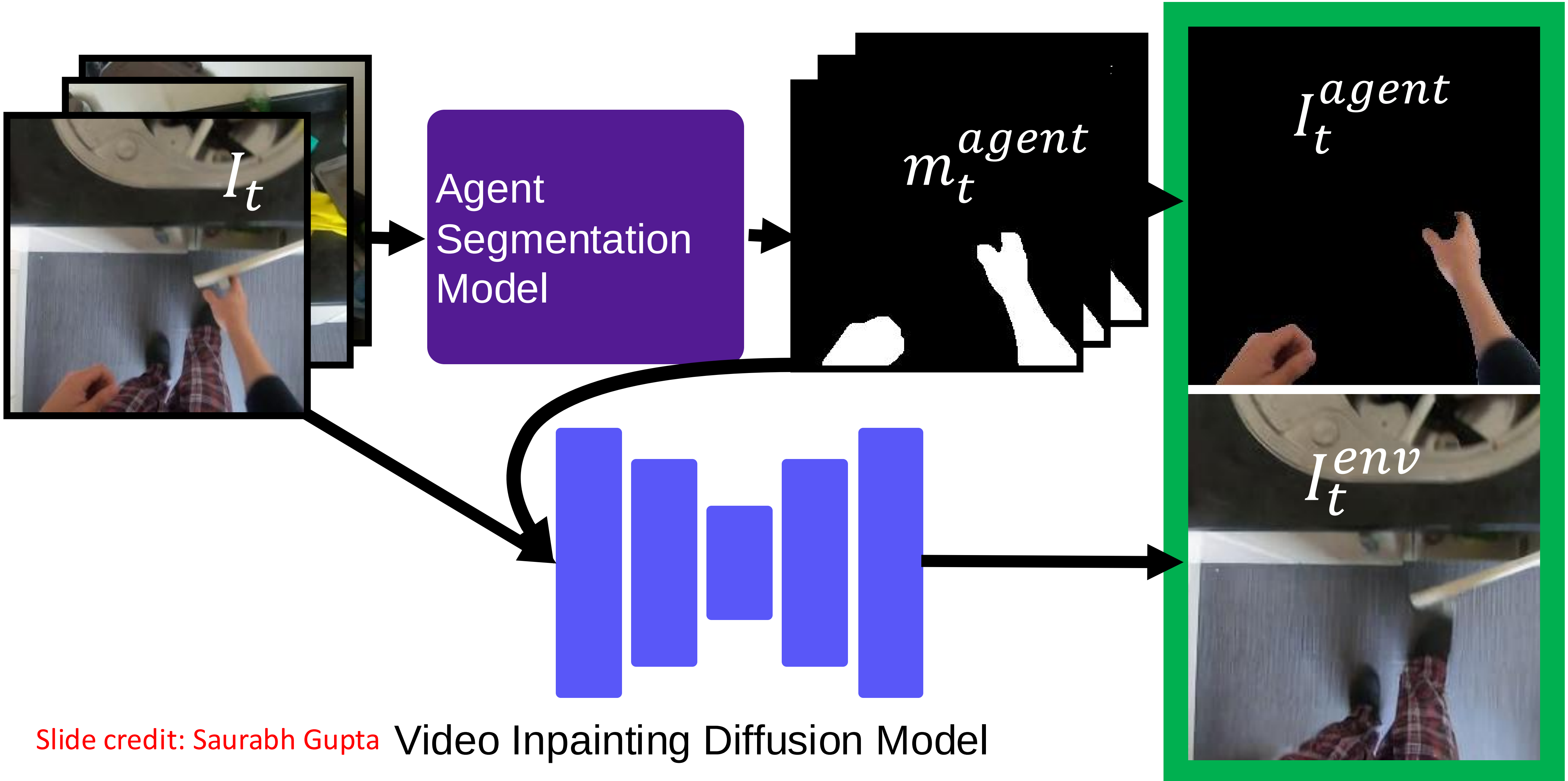


Visualizations

Visualizations

Visualizations

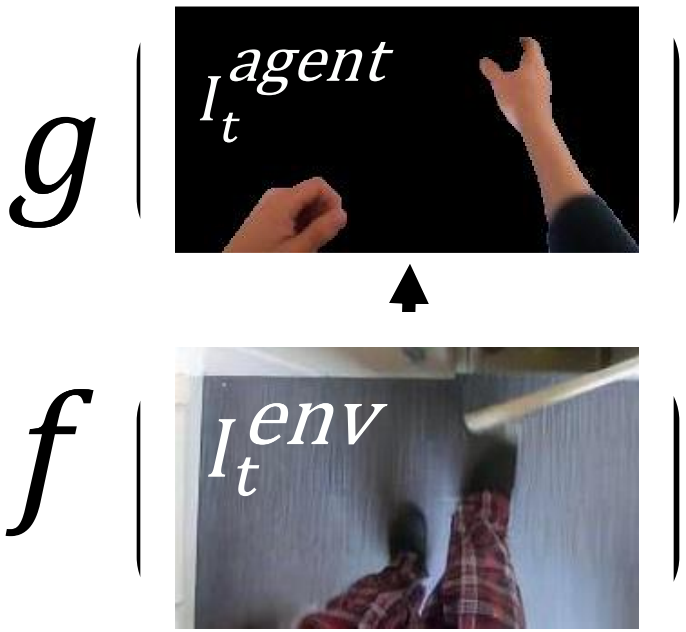
Agent-Environment *Factored* Representations



Slide credit: Saurabh Gupta

Using the *Factored* Representations

a) Affordances



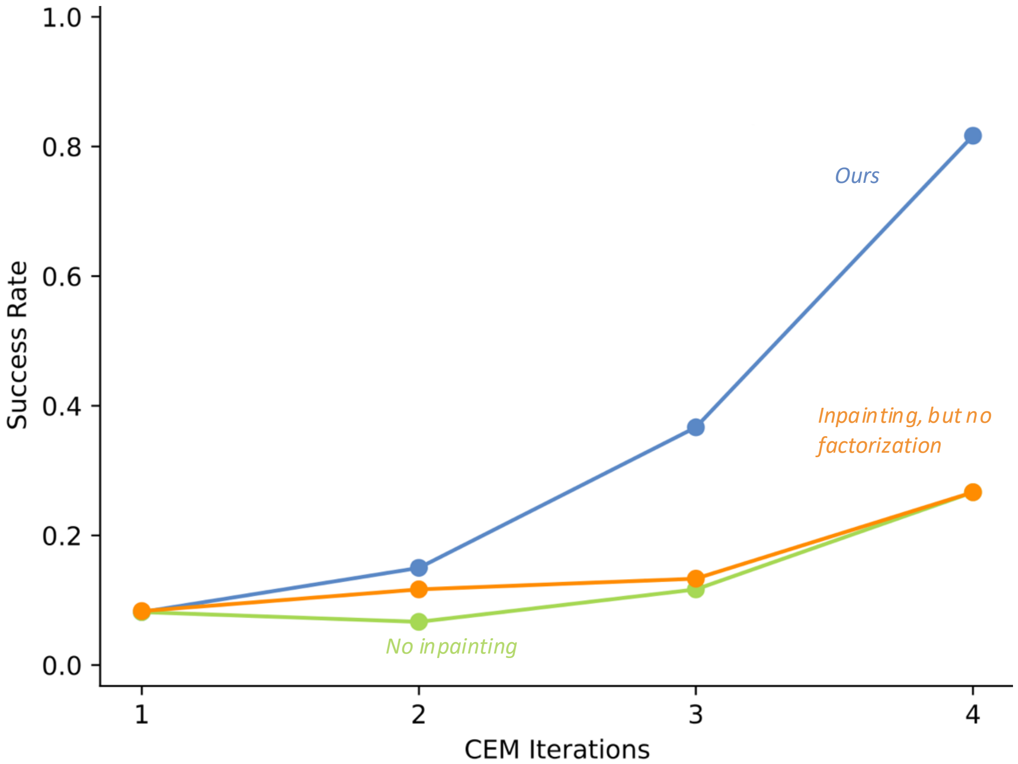
b) Reward functions from human data

Learn reward function from human data ⇒ Use it to train robots

Slide credit: Saurabh Gupta

Robot Learning through Reward Functions Learned on Video Data

Slide credit: Saurabh Gupta



Affordances from Human Videos as a Versatile Representation for Robotics

Shikhar Bahl*[1,2] Russell Mendonca*[1] Lili Chen[1] Unnat Jain[1,2] Deepak Pathak[1]

[1]CMU [2]Meta AI

Figure 1. We leverage human videos to learn visual affordances that can be deployed on multiple real robot, in the wild, spanning several tasks and learning paradigms. Videos available at https://vision-robotics-bridge.github.io/.

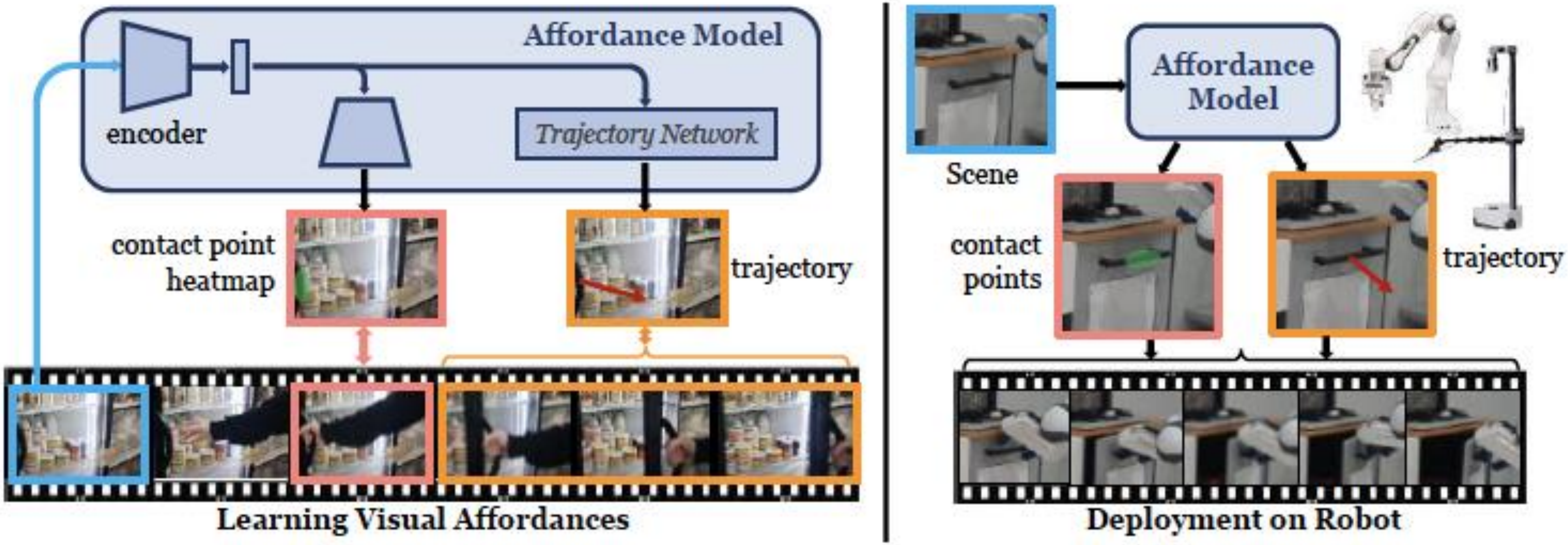


Figure 2. **VRB Overview**. First, we learn an actionable representation of visual affordances from human videos: the model predicts contact points and trajectory waypoints with supervision from future frames. For robot deployment, we query the affordance model and convert its outputs to 3D actions to execute.

https://vision-robotics-bridge.github.io/

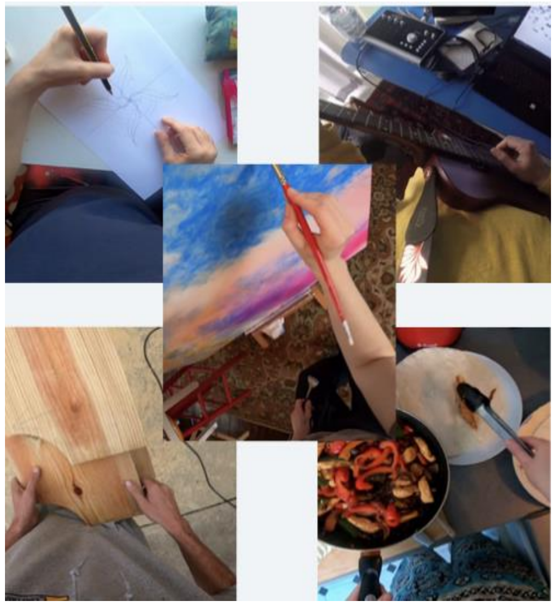
Reconstructing 4D Hand-Object Interactions from Videos

Hand-Object Interaction Pretraining from Videos

Himanshu Gaurav Singh*, Antonio Loquercio*,
Carlo Sferrazza, Jane Wu, Haozhi Qi, Pieter Abbeel, Jitendra Malik

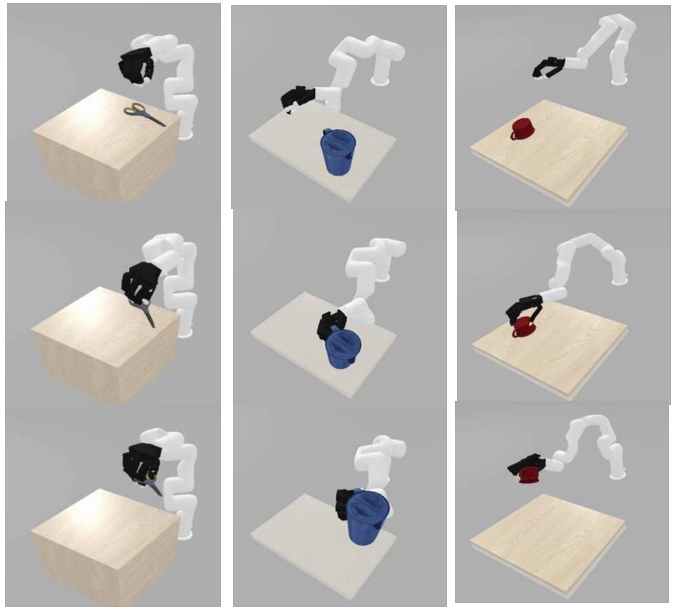
Robot data from videos

Video Dataset



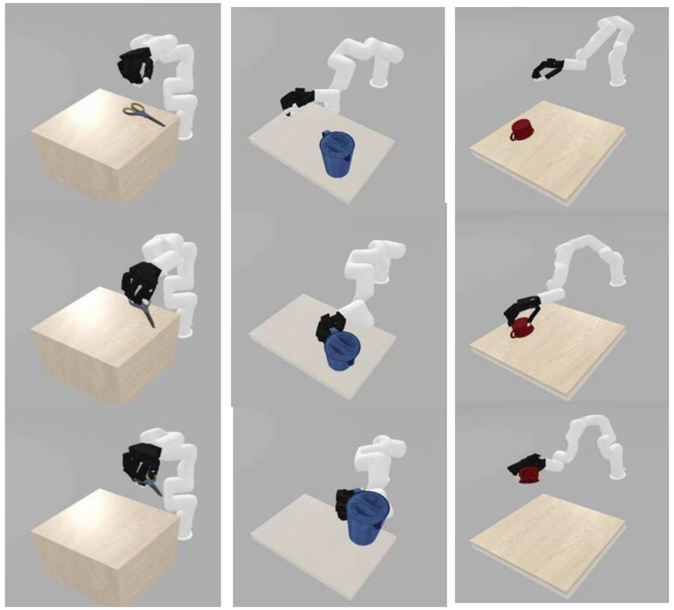
3-D hand object trajectories

Robot Dataset

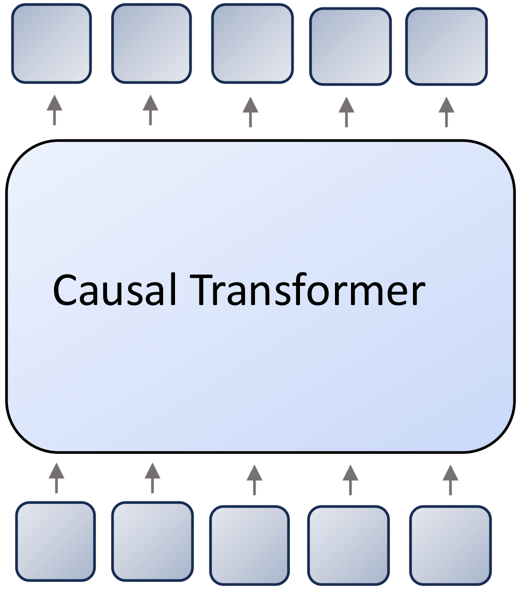


Learning robot policies

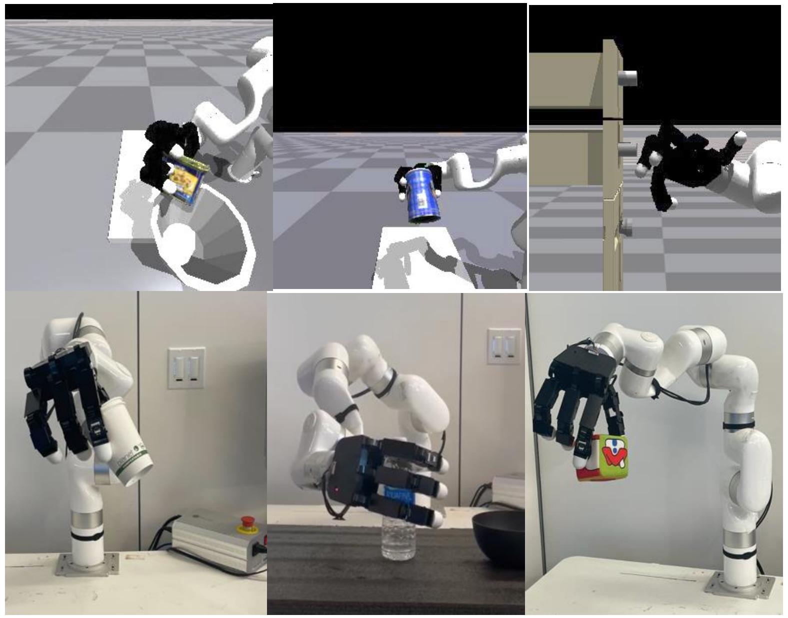
Robot Dataset



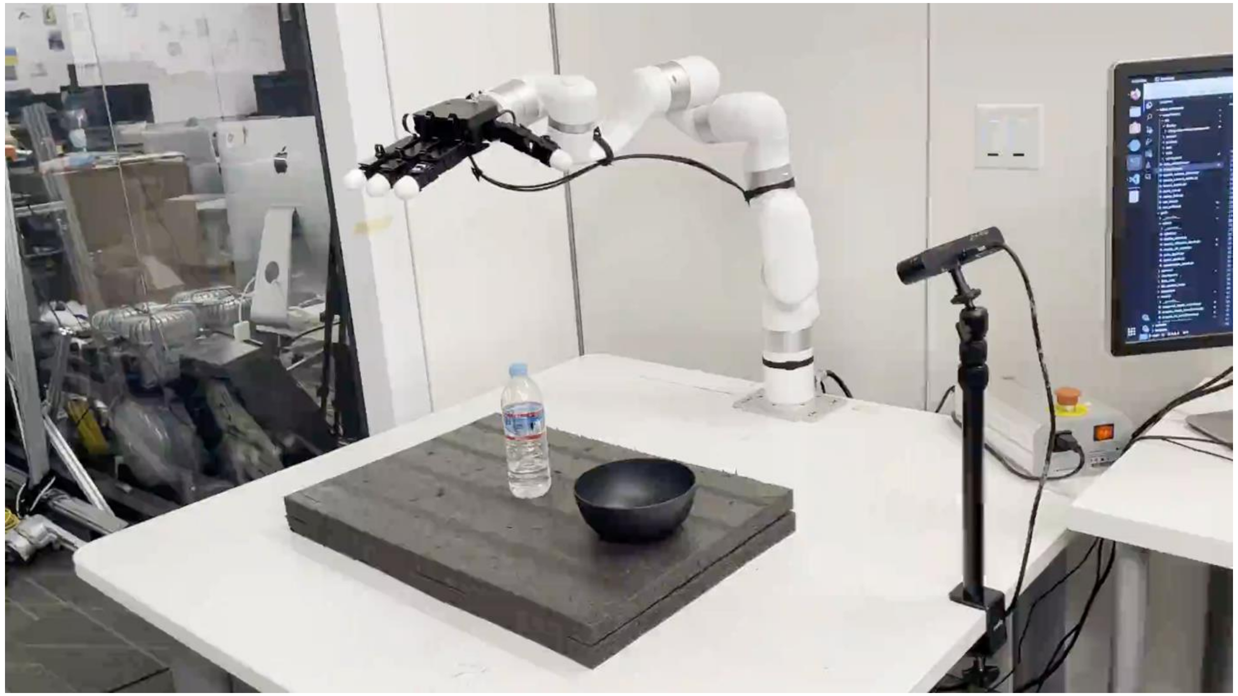
General Manipulation Prior



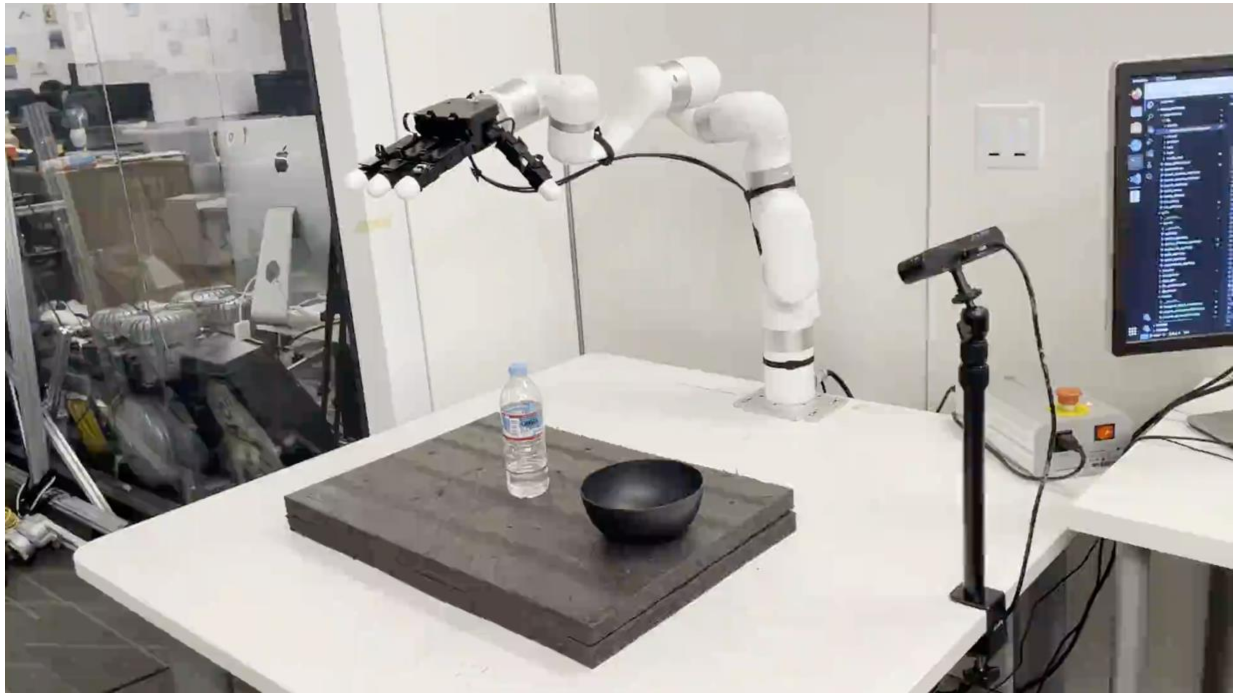
Downstream adaptation



Grasp and Pour



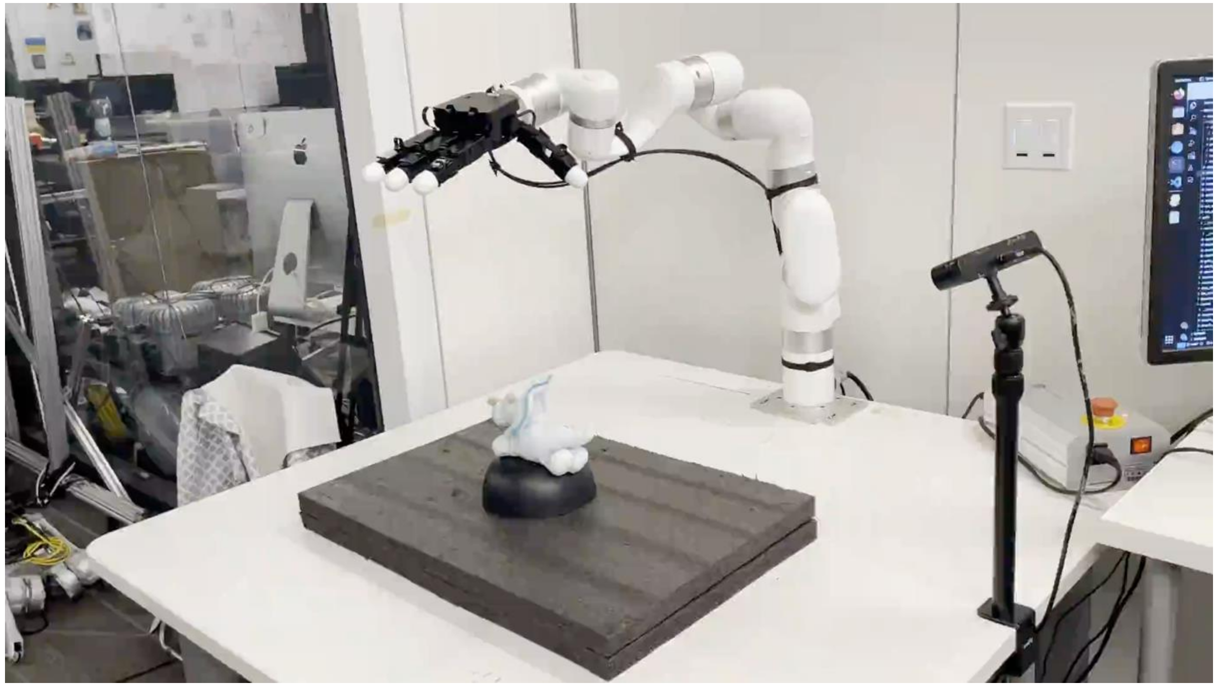
Ours



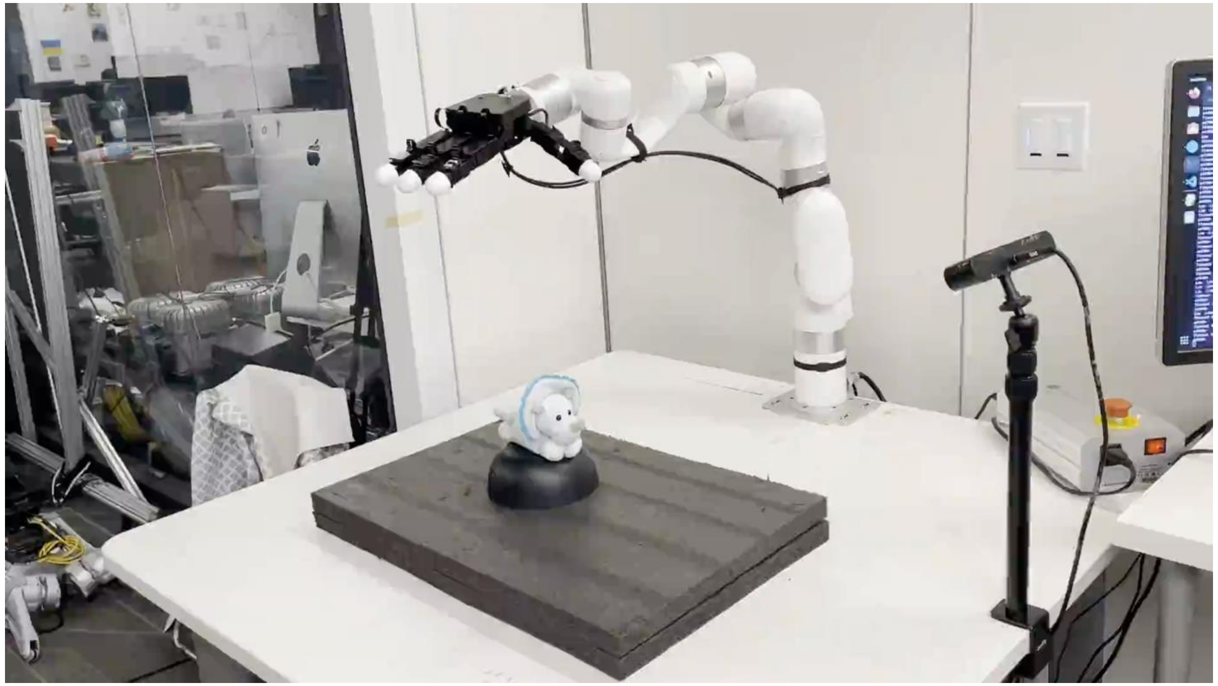
Diffusion Policy
(shaky trajectory)

After downstream finetuning with < 15 demonstrations

Grasp and Lift (toy)



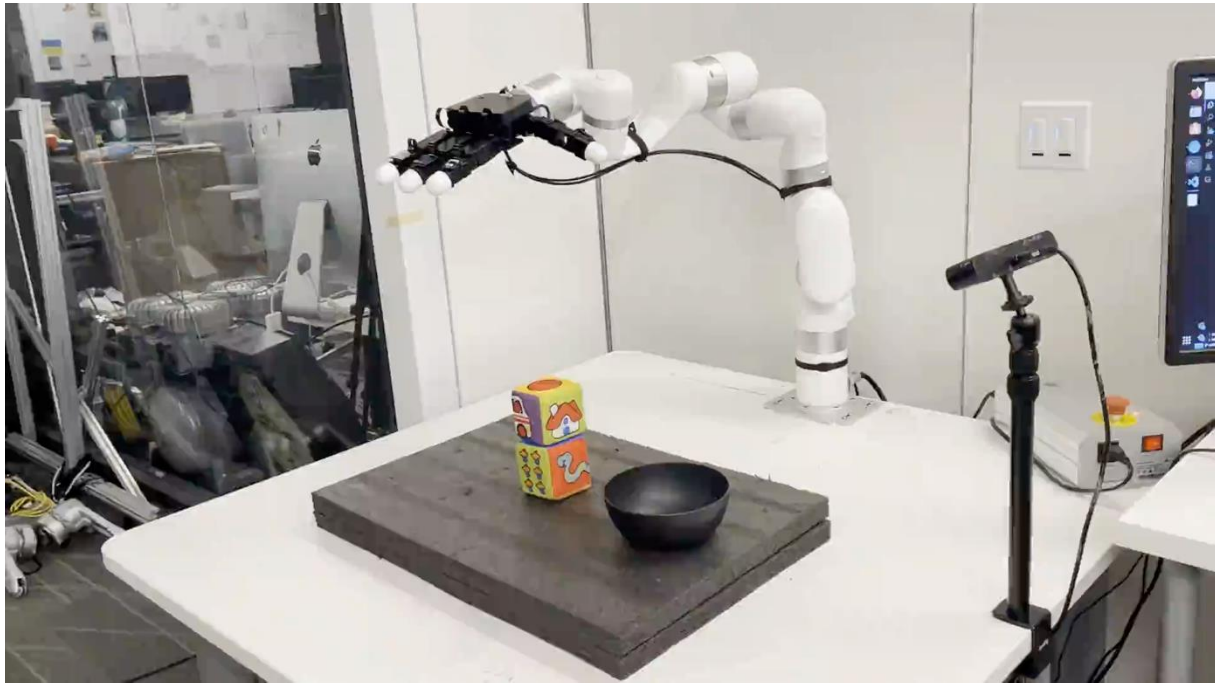
Ours



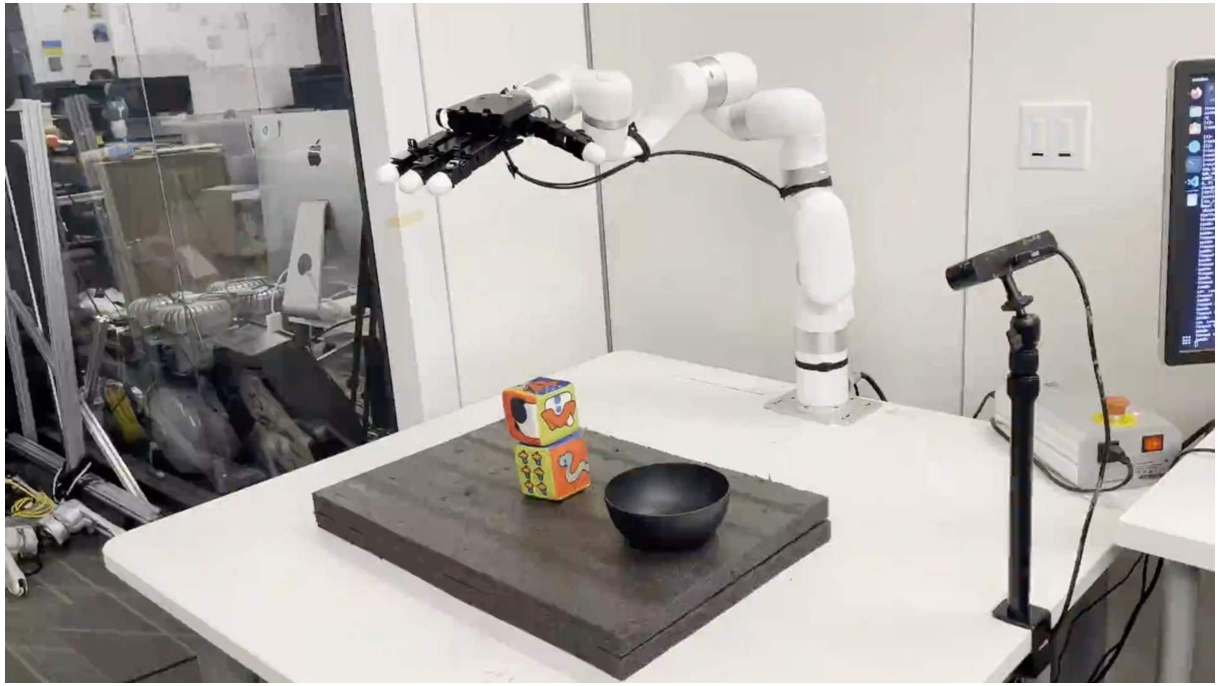
Diffusion Policy
(shaky trajectory)

After downstream finetuning

Grasp and Throw (cube)



Ours



Diffusion Policy
(shaky behavior)

After downstream adaptation with < 15 demonstrations

Thank you!